Analytics & Big Data

Session 10: Analytics in organizations

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🎯 Learning objectives

- **Explain** the strategic role of data analytics in data-driven organizations.
- **Discuss** the ethical and legal boundaries of data analytics in organizations.

Level: Industry

Does the deployment of analytics pay off?

The Effect of Big Data and Analytics on Firm Performance: An Econometric Analysis Considering Industry Characteristics

Müller et al. (2018)

- objective measures (specifically: TODO)
- 800 companies

$$\begin{aligned}\log(Sales) = \beta_0 &+ \beta_1 \log(Labor) + \beta_2 \log(Capital) + \beta_3 \log(Materials) \\ &+ \beta_4 BDA + Controls + \varepsilon\end{aligned}$$

Findings

	OLS (1)	FE (2)
log(Capital)	0.090*** (0.015)	0.127*** (0.029)
log(Labor)	0.296*** (0.024)	0.472*** (0.043)
log(Materials)	0.667*** (0.023)	0.442*** (0.035)
BDA	0.041** (0.020)	0.038* (0.021)
Industry dummies?	Yes	Yes
Year dummies?	Yes	Yes
IT Asset dummies?	Yes	Yes
Observations	5,698	5,698
R^2	0.977	0.791
Adjusted R^2	0.977	0.755

Note: highlight:

- methods (OLS + fixed effects/causality)
- differences between industries (context matters)

TODO: why is R^2 so high?!?

“Competing on analytics”

Level: Organization

Under which conditions could we expect higher performance?

Oesterreich et al. (2022)

TODO: details (literature review, 125 studies, …)

TBD: prepare/anticipate the connection from north-star KPI to quantitative metrics and analytical models?

Meta-analysis results

TODO: check table!

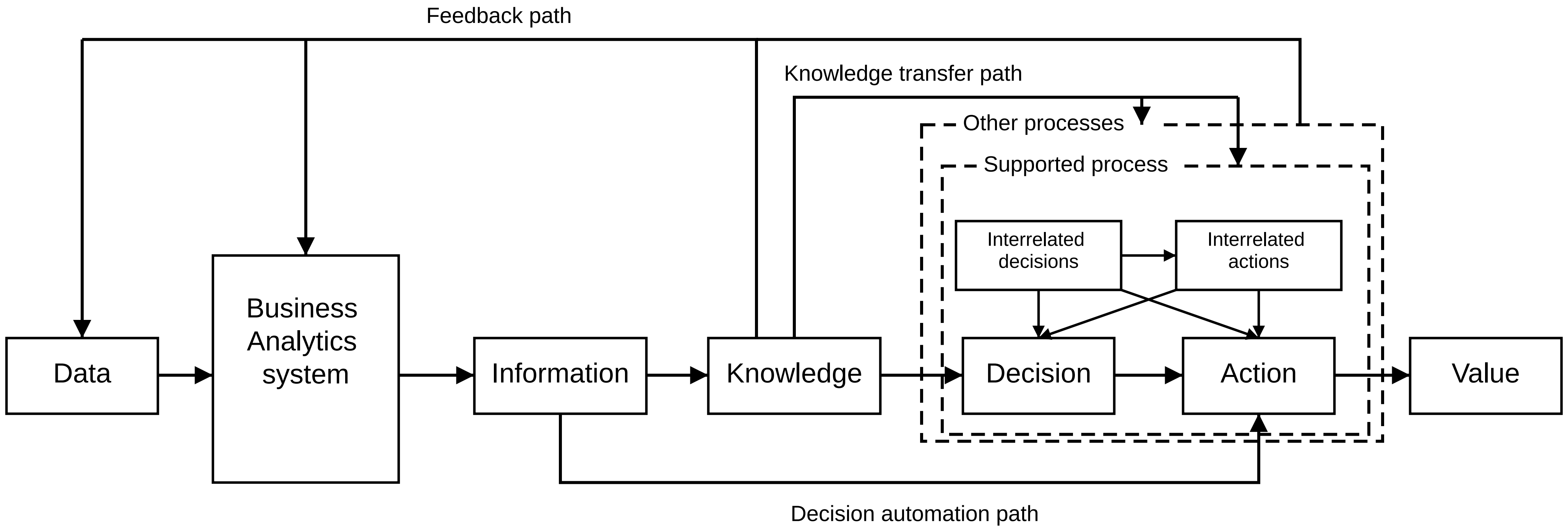
Independent variable	n	N	Estimated effect	95% CI	Heterogeneity
BA technical resources	91	39,246	0.452	0.409–0.493	$Q = 1920.153^{**}$, $df = 90$, $I^2 = 95.31\%$
BA human resources	42	9,219	0.494	0.444–0.541	$Q = 379.967^{**}$, $df = 41$, $I^2 = 89.21\%$
BA management capabilities	69	16,074	0.477	0.433–0.519	$Q = 861.169^{**}$, $df = 68$, $I^2 = 92.10\%$
Data quality	14	3,088	0.434	0.328–0.528	$Q = 142.257^{**}$, $df = 13$, $I^2 = 90.86\%$
Organizational culture	18	3,698	0.472	0.391–0.546	$Q = 153.939^{**}$, $df = 17$, $I^2 = 88.96\%$
External pressure	17	3,644	0.165	0.036–0.288	$Q = 248.937^{**}$, $df = 16$, $I^2 = 93.57\%$

Level: Process

TODO

Kunz et al. (2025)

Value Creation Paths from BA



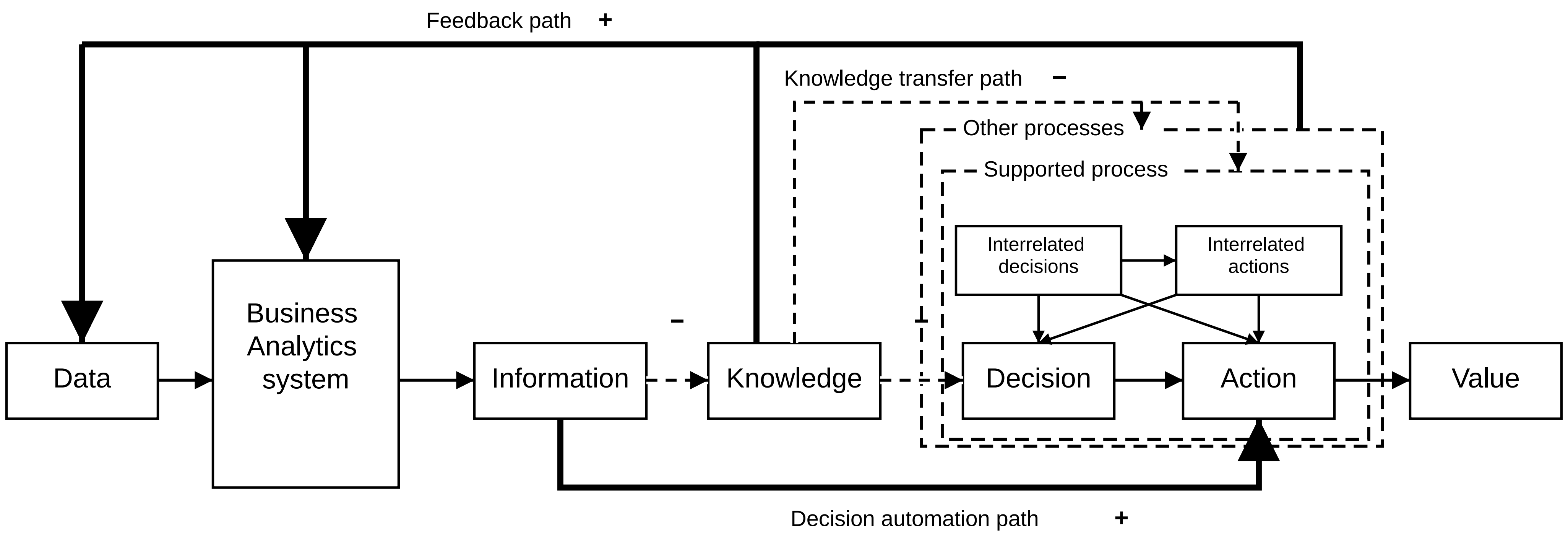
Note: maybe add explanations on the following slide?!

TODO:

- cover canonical information processing path, decision augmentation vs. automation
- cover knowledge accumulation path, transfer, feedback

TODO: summarize each value creation path (autonomy, dynamism, opacity)

Changes (ML)



Feedback path +
Knowledge transfer path –
Other processes
Supported process
Interrelated decisions
Interrelated actions
Data
Business Analytics system
Information
–
Knowledge
Decision
Action
Value
Decision automation path +

Level: Human-AI interaction

Skills and adoption (perception??)

Human-AI collaboration

Autonomy vs.

Collaboration modality (ghostwriter vs. sounding board) interact with expertise and determine quality of creative knowledge work ([[ChenChan2024]]): using [[LLM]]s for ghostwriting is detrimental to experts, as a sounding board, it helps non-experts.

When delegating tasks (such as classification), humans may be less effective than AI (see [[FugenerGrahlGuptaEtAl2022]])

Understanding the [[Jagged Technological Frontier]] is challenging -> maybe, humans are no longer in a position to decide (in some cases)

Proactive agents lead to a higher loss of users' competence-based self-esteem and reduce system satisfaction (compared to reactive agents) ([[DiebelGoutierAdamEtAl2025]])

Roles interactions: algorithms in their multiple roles as supervisors/evaluators/peers create role conflict and role ambiguity, which prompts humans to react (specific tasks or cognitive reactions), feeding into the *computational feedback learning loop* and/or the *(broken) human feedback loop* [[TarafdarPageMarabelli2023]]

Human: Strich/JAIS -> "safeguarding against human intervention" (loan agents cannot overrule the AI decision, preventing aself-interest to affect their decisions)

Boundaries

TBD. but students should have an overview here

XAI (with illustrations, as a legal requirement)

TBD: situational overrides (emergencies?)

Summary

- TODO

Survey: Session 10

https://forms.gle/Rhr4C47jsAZJyumN6

Note: Responses may be analyzed and published in anonymized form.

Please complete the survey before you leave today — thank you 🙏

References

Kunz, P. C., Spohrer, K., & Heinzl, A. (2025). Process-level value creation from business analytics: A theoretical literature review of value creation paths and changes induced by machine learning. *The Journal of Strategic Information Systems*, *34*(2), 101902. https://doi.org/10.1016/j.jsis.2025.101902

Müller, O., Fay, M., & Vom Brocke, J. (2018). The effect of big data and analytics on firm performance: An econometric analysis considering industry characteristics. *Journal of Management Information Systems*, *35*(2), 488–509. https://doi.org/10.1080/07421222.2018.1451955

Oesterreich, T. D., Anton, E., & Teuteberg, F. (2022). What translates big data into business value? A meta-analysis of the impacts of business analytics on firm performance. *Information & Management*, *59*(6), 103685. https://doi.org/10.1016/j.im.2022.103685